会算术的马

The Horse That Could Count



汉斯是在算数，还是在读懂人的动作？

英文原文

Wilhelm von Osten, a mathematics teacher and **trainer** in Berlin, owned a **stallion**, Hans. Asked to **calculate** a sum, attempt **subtraction**, or name a date, Hans tapped out the answer with a **hoof**. Such **aptitude** suggested **brilliance**, and **demonstrations** in his owner's **courtyard** made him seem **intelligent**. The New York Times reported the **stunning** act in 1904.

A **commission** of thirteen was formed to **investigate**. It included a **veterinarian**, a circus manager, a **cavalry** officer, schoolteachers, and the director of the Berlin **zoological** garden. Its members looked for **trickery** and **collusion**. In September 1904 they found no **deception**, in effect **certifying** the act **authentic**.

In 1907 the **psychologist** Oskar Pfungst tested the act **systematically**, altering one **variable** after another. He kept onlookers away, **rotated questioners**, fitted **blinders**, and sometimes used people **ignorant** of the answer. When Hans could see the questioner, he answered **correctly** 50 times in 56. With the questioner **invisible**, his **accuracy** fell to two in 35.

The **cues** came from the questioners. Each would **lean** forward as tapping began, and **tension** rose as the taps neared the answer. At the right tap it eased, and the questioner would **straighten** or **jerk** the head upward. Every **gesture** and **motion** was **subconscious**, made **unwittingly** by people with no wish to **deceive**.

Double-blind designs in **psychology** and animal **cognition guard against** the Clever Hans effect, a **phenomenon** of **bias** in which **expectancy** shapes **behavioural** results. In the laboratory, Pfungst took the animal's place and proved **sensitive** to the cues of **attentive** volunteers. Hans went, on his owner's death in June 1909, to a **jeweller** still convinced that such animals **reasoned**.

中文译文

威廉·冯·奥斯滕，一位数学教师兼驯兽师在柏林（任职），拥有一匹公马，名叫汉斯。当（这匹马）被要求计算一道算术题、试着做减法或说出一个日期时，汉斯就会用一只蹄子把答案敲出来。这样的天资暗示着过人的才智，在它主人庭院里的演示/展示也让它显得很聪明。《纽约时报》在1904年报道了这场令人惊叹的表演。

一个由十三人组成的委员会成立了，负责调查。委员会成员包括一名兽医、一名马戏团经理、一名骑兵军官、几名中小学教师以及柏林动物园的园长。委员们查看有没有诡计和串通。1904年9月，他们没有发现任何骗局，这实际上等于证实了这场表演真实可信。

1907年，心理学家奥斯卡·普丰斯特系统地检验了这场表演，一个接一个地改变了变量。他不让围观者靠近，轮换提问者，（给马）戴上眼罩，有时还使用对答案一无所知的人（当提问者）。当汉斯能看见提问者时，它在56次里有50次正确地作答。当提问者不可见时，它的准确率降到了35次里只对两次。

（看来）这些提示来自提问者。每个提问者都会在敲击开始时向前倾身，敲击越接近答案，紧张感就越强。敲到正确的那一下时，紧张感就缓和下来，提问者会挺直身子或者把头向上猛地一抬。每一个姿势和动作都是下意识的，是那些并不想欺骗别人的人毫不知情地做出的。

双盲设计在心理学和动物认知研究中用来防范聪明汉斯效应，这是一种偏差现象，在这种现象中，预期会影响行为结果。在实验室里，普丰斯特代替了那匹马，结果证明，他对专注的志愿者（无意中）的提示也很敏感。汉斯在它的主人于1909年6月去世后，归了一位珠宝商，这位珠宝商仍然坚信这类动物会推理。

词汇与表达

trainer n.
驯兽师

stallion n.
公马

calculate v.
计算

subtraction n.
减法

hoof n.
蹄子
the answer with a hoof.

aptitude n.
天资
Such aptitude suggested brilliance,

brilliance n.
过人的才智

demonstrations n.
演示/展示

courtyard n.
庭院

intelligent adj.
聪明

stunning adj.
令人惊叹的

commission n.
委员会
A commission of thirteen

investigate v.
调查
to investigate.

veterinarian n.
兽医

cavalry n.
骑兵

zoological adj.
动物

trickery n.
诡计

collusion n.
串通
trickery and collusion

deception n.
骗局
they found no deception

certifying v.
证实了
certifying the act authentic

authentic adj.
真实可信
certifying the act authentic

psychologist n.
心理学家

systematically adv.
系统地
tested the act systematically

variable n.
变量
altering one variable after another.

rotated v.
轮换

questioners n.
提问者

blinders n.
眼罩

ignorant adj.
一无所知的

词汇与表达（续）

correctly adv.
正确地

invisible adj.
不可见

accuracy n.
准确率
his accuracy fell to two in 35.

cues n.
提示
The cues came from the questioners.

lean v.
倾身

tension n.
紧张感
and tension rose

straighten v.
挺直

jerk v.
猛地一抬

gesture n.
姿势
Every gesture and motion

motion n.
动作

subconscious adj.
下意识的
motion was subconscious

unwittingly adv.
毫不知情地
made unwittingly by people

deceive v.
欺骗
no wish to deceive

Double-blind adj.
双盲
Double-blind designs

psychology n.
心理学

cognition n.
认知
animal cognition

guard against phr.
防范
animal cognition guard against the Clever Hans effect,

phenomenon n.
现象
a phenomenon of bias

bias n.
偏差
a phenomenon of bias

expectancy n.
预期
expectancy shapes behavioural results.

behavioural adj.
行为

sensitive adj.
敏感
proved sensitive to the cues

attentive adj.
专注的

jeweller n.
珠宝商

reasoned v.
推理
that such animals reasoned.

文章简述

- 奇观：柏林一匹名叫汉斯的马用蹄子敲出算术答案；早期调查没发现作弊，人们便更相信它会计算。
- 检验：心理学家普丰斯特改变提问和视线条件：汉斯看得见提问者时，56次答对50次；看不见时，35次仅对2次。
- 信号：知道答案的提问者，在马敲到正确次数时会不自觉地放松、挺身或抬头；汉斯察觉变化就停下。提问者并不知道自己在提示，也没有故意骗人。
- 启示：普丰斯特自己扮马，也能读出志愿者的提示；人的期待可能无意中影响实验。双盲设计让实验者和受试者都不知道分组或答案，用来减少这种影响。
- 余波：汉斯的主人去世后，马归了一位珠宝商；尽管实验已揭示身体信号的作用，珠宝商仍坚信这类动物会推理。